PQC-FL-Health: Post-Quantum Federated Learning for Privacy-Preserving Healthcare Analytics

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***Abstract*—Federated learning (FL) allows hospitals to collaboratively train machine learning models without transferring raw patient data. Most existing FL deployments use classical cryptographic schemes such as RSA and HMAC, which are vulnerable to harvest-now-decrypt-later attacks by quantum-capable adversaries. We present PQC-FL-Health, a three-layer security framework for federated healthcare analytics that replaces classical primitives with NIST-standardised post-quantum cryptography (PQC): (1) ML-KEM-768 (CRYSTALS-Kyber) for quantum-safe key encapsulation, (2) ML-DSA-65 (CRYSTALS-Dilithium) for authenticated gradient updates, and (3) Gaussian differential privacy (DP) to guard against aggregator-side patient-data inference. We evaluate the system on a node-specialised five-hospital federation in which each hospital holds its own disease domain (Pima diabetes, Cleveland heart disease, WDBC breast cancer), and on a homogeneous single-disease scenario with non-IID glucose-quartile partitioning as the primary FL evaluation. Over five random seeds and 50 communication rounds, PQC-FL-Health achieves global AUC 0.888 ± 0.005 on the heterogeneous federation, with per-disease AUC of 0.798 (diabetes), 0.845 (heart disease), and 0.996 (breast cancer). PQC operations complete in under 1 ms at all three NIST security levels and run 1.6× faster end-to-end than RSA-4096 (64 ms vs. 103 ms per round), despite adding 31.1 KB of cryptographic overhead per round. ML-DSA signature verification detects 100% of tampered-signature attacks; a gradient-norm filter catches all tested scaled-gradient poisoning attempts. At $\varepsilon = 10$, differential privacy costs only 4.4% AUC, while budgets at or below $\varepsilon = 1$ cause severe degradation due to noise overwhelming the gradient signal in a small model. All code, datasets, and results are publicly available.**

***Index Terms*—federated learning, post-quantum cryptography, differential privacy, Byzantine robustness, ML-KEM, ML-DSA, healthcare AI, CRYSTALS-Kyber, CRYSTALS-Dilithium**

I. INTRODUCTION

Healthcare institutions generate large volumes of sensitive patient data that are distributed across hospitals, clinics, and research centres. Federated learning (FL) [1] lets these institutions jointly train shared models without centralising raw records, preserving patient privacy by design. However, FL deployments in production healthcare systems face two threats that are rarely addressed together.

The quantum threat. Classical public-key cryptography, including RSA, Diffie-Hellman, and ECDSA, underpins the secure channels over which FL updates travel. Shor's algorithm [2] can break these schemes in polynomial time on a sufficiently powerful quantum computer. More immediately, adversaries can record encrypted model updates today and decrypt them once quantum hardware matures, exposing both the model and the patient data used to train it. NIST has responded by standardising ML-KEM (FIPS 203) [3] and ML-DSA (FIPS 204) [4], two lattice-based schemes designed to resist quantum attacks.

The federated attack surface. Even with a secure channel, FL aggregation is vulnerable to Byzantine nodes that inject malicious gradient updates [5], and to honest-but-curious aggregators that infer patient attributes from received updates [11]. Differential privacy (DP) [6] is the standard defence against aggregator inference, but its interaction with FL convergence under non-IID healthcare data is not well characterised empirically.

Contributions. This paper makes the following contributions:

- We design and implement **PQC-FL-Health**, a federated learning system for healthcare that, to the best of our knowledge, is the first to simultaneously provide quantum-safe channel security (ML-KEM-768), authenticated updates (ML-DSA-65), Byzantine robustness via signature verification and gradient-norm filtering, and differential privacy via the Gaussian mechanism with empirical benchmarks against RSA-4096.
- We benchmark all three NIST post-quantum security levels (ML-KEM-512/768/1024 paired with ML-DSA-44/65/87) on a realistic 12 KB model-delta payload, reporting latency, bandwidth overhead, and convergence quality.
- We evaluate two clinically motivated federation scenarios: a node-specialised five-hospital federation spanning three disease domains with local-vs-federated comparison, and a homogeneous five-clinic federation for diabetes prediction under extreme non-IID glucose-quartile partitioning.
- We quantify the privacy-utility tradeoff for DP in small-model FL, showing that $\varepsilon \geq 10$ costs less than 5% AUC, while $\varepsilon \leq 1$ is not viable for a 1531-parameter model, providing a practical baseline for future work on DP-compatible architectures.

Section II reviews related work. Section III describes the system architecture. Section IV covers the experimental setup. Section V presents results. Section VI concludes.

II. RELATED WORK

FedAvg [1] remains the dominant FL aggregation rule; Li et al. [12] survey its challenges including non-IID data [14] and systems heterogeneity. FL in medical imaging [13] motivates privacy-preserving aggregation without data centralisation. PQC for FL is nascent: Cheng et al. [15] propose a PQC communication model for quantum FL; Huang et al. [16] study NIST PQC signatures in FL; PQS-BFL [17] combines PQC with blockchain-based FL; a 2025 survey [18] reviews the FL–quantum intersection. On Byzantine robustness, Blanchard et al. [5] introduced Krum; Fang et al. [10] showed adaptive poisoning can bypass many defences. On privacy, Abadi et al. [7] introduced DP-SGD; Geyer et al. [8] applied client-level DP to FL. To the best of our knowledge, no prior work combines ML-KEM key encapsulation, ML-DSA gradient authentication, Byzantine defence-in-depth, and Gaussian DP in a single healthcare FL system with empirical benchmarks against RSA-4096.

III. SYSTEM DESIGN

A. Architecture Overview

PQC-FL-Health uses a standard cross-silo FL topology: $N = 5$ hospital nodes communicate with a central aggregation server over simulated secure channels. Each node holds a private local dataset and a local model copy. In each communication round, nodes: (1) receive the current global weights from the server, (2) train locally for $E = 5$ epochs, (3) compute a weight delta $\Delta_i = w_i^{\text{new}} - w^{\text{global}}$, (4) optionally apply DP noise, (5) encrypt and sign Δ_i using PQC primitives, and (6) transmit the bundle. The server verifies each bundle, aggregates accepted updates with FedAvg, and broadcasts the new global weights.

B. Layer 1: Quantum-Safe Confidentiality (ML-KEM-768)

The server generates a long-term ML-KEM-768 keypair and publishes its public key. Before sending an update, each node encapsulates a shared secret:

$$(\text{ct}, K) \leftarrow \text{ML-KEM.Encaps}(\text{pk}_{\text{server}}) \tag{1}$$

The shared secret K is expanded via HKDF-SHA256 to a 256-bit AES-GCM key, which encrypts the serialised gradient delta:

$$c \leftarrow \text{AES-256-GCM.Enc}(K,\ \Delta_i) \tag{2}$$

The transmitted bundle contains (ct, c, nonce). The server decapsulates ct to recover K and decrypts c. ML-KEM-768 achieves NIST security level 3 (AES-192 equivalent) with a ciphertext of 1088 bytes and average encapsulation latency under 0.3 ms in our environment.

C. Layer 2: Authenticated Updates (ML-DSA-65)

Each node holds a long-term ML-DSA-65 signing keypair; the server maintains a registry of node public keys established at setup time and not retransmitted each round. After encryption, the node signs the concatenation of the KEM ciphertext and AES ciphertext:

$$\sigma \leftarrow \text{ML-DSA.Sign}(\text{sk}_i,\ \text{ct}\|c) \tag{3}$$

The server verifies σ before accepting any update. A node that sends a tampered or forged signature is immediately rejected and its update discarded. ML-DSA-65 signatures are 3309 bytes with verification latency under 0.3 ms, providing NIST security level 3 authentication.

D. Layer 3: Aggregator-Side Privacy (Differential Privacy)

To protect against an honest-but-curious aggregator that may infer patient attributes from gradient updates, we apply the Gaussian mechanism [6] at the node level. Each node clips its delta to ℓ_2 sensitivity $S = 0.1$ and adds calibrated Gaussian noise:

$$\tilde{\Delta}_i = \text{clip}_S(\Delta_i) + \mathcal{N}(0,\ \sigma^2\mathbf{I}), \quad \sigma = \frac{S\sqrt{2\ln(1.25/\delta)}}{\varepsilon} \tag{4}$$

with $\delta = 10^{-5}$ fixed and ε as the privacy budget. DP noise is added before PQC encryption, so the server never observes the original gradient. Privacy loss is tracked across rounds using Rényi differential privacy (RDP) composition [9], which yields tighter cumulative bounds than naive linear composition; at 50 rounds and $\varepsilon_{\text{per-round}} = 10$, the RDP accountant gives $\varepsilon_{\text{cumul}} = 181.7$ vs. 500 under naive addition.

E. Byzantine Defence

Our defence operates at two levels. First, ML-DSA signature verification (Layer 2) immediately rejects any bundle that fails cryptographic authentication, covering tampered-signature attacks with 100% detection. Second, after decryption, the server optionally applies a gradient-norm anomaly filter:

$$\text{reject node } i \text{ if } \|\Delta_i\|_2 > 3 \cdot \text{median}_j \|\Delta_j\|_2 \tag{5}$$

This catches gradient-scaling attacks where a rogue node sends a correctly signed but amplified delta.

F. Aggregation

Accepted updates are aggregated with FedAvg weighted by local dataset size:

$$w^{\text{new}} = w^{\text{global}} + \sum_{i\in\mathcal{A}} \frac{n_i}{\sum_{j\in\mathcal{A}} n_j}\,\Delta_i \tag{6}$$

where $\mathcal{A}$ is the set of accepted nodes and n_i is node i's training set size.

IV. EXPERIMENTAL SETUP

A. Datasets and Non-IID Partitioning

We evaluate on two federation scenarios.

Homogeneous federation (FL convergence benchmark). All five nodes hold disjoint shards of the Pima Indians Diabetes dataset [20] (768 samples, 8 features, 34.9% positive), partitioned *non-IID* by ascending glucose level (feature 2). This produces extreme label skew: node 0 (low-glucose patients) has 8.1% positive rate, while node 4 (high-glucose referral centre) has 73.2% positive rate, a 9× ratio. This is a methodologically clean cross-silo FL benchmark: all nodes share the same prediction task and label semantics.

Heterogeneous federation (primary evaluation). Five simulated hospitals each specialise in a single disease domain: nodes 0–1 hold independent shards of the Pima Indians Diabetes dataset [20] (8 features, ∼205 training samples each); node 2 holds the Cleveland Heart Disease dataset [21] (13 features, 238 training samples, 46% positive); node 3 holds the Wisconsin Diagnostic Breast Cancer (WDBC) dataset [22] (30 features, 456 training samples, 37% positive); node 4 holds a third Pima shard with Gaussian feature noise ($\sigma = 0.1$) simulating a lower-quality data source. All features are zero-padded to 30 dimensions. Each hospital normalises its own features independently. This scenario tests whether PQC-FL-Health operates correctly under data heterogeneity; a local-vs.-federated comparison (Section V) quantifies the benefit and cost of cross-disease gradient sharing.

B. Model and Training

Each hospital trains a fully connected multi-layer perceptron (MLP) with three layers ($30 \to 64 \to 32 \to 1$, ReLU activations, sigmoid output, 1531 parameters) for $E = 5$ local epochs per round using SGD with learning rate $\eta = 0.01$. An MLP is appropriate here given that all three datasets are tabular. Binary cross-entropy loss is used throughout with mini-batches of size 32.

C. Evaluation Protocol

We run 50 communication rounds per seed across five random seeds (42, 123, 456, 789, 1337) and report mean ± standard deviation. The global test set is the union of all hospital test sets; for the heterogeneous scenario we also report per-disease AUC separately. The primary metric is AUROC (AUC); we also report sensitivity, specificity, F1, and accuracy.

Cryptographic benchmarks use 20 iterations per configuration on a 12 KB payload representative of a model delta. All experiments run on Python 3.12 under Ubuntu WSL2 (Linux 6.6, no GPU) on an AMD Ryzen 7 7435HS CPU, using liboqs 0.15.0 via liboqs-python 0.14.1.

V. Results

A. FL Convergence and Local Baseline

Fig. 1 shows convergence curves for both federation scenarios across five seeds.

Homogeneous Pima-only federation (primary). Final AUC is 0.787 ± 0.019 (Table I). The extreme non-IID skew—nodes spanning 8–73% positive rate—forces FedAvg to reconcile fundamentally different local optima, explaining the residual variance and the gap relative to a balanced dataset.

Heterogeneous federation (system stress-test). Global AUC rises from 0.472 ± 0.082 (round 1) to $\mathbf{0.888 \pm 0.005}$ (round 50). Fig. 1 (left) also shows per-disease convergence: breast cancer (WDBC) saturates early at AUC 0.996 ± 0.002, while diabetes and heart disease converge more gradually to AUC 0.798 ± 0.013 and 0.845 ± 0.011, respectively (Table I).

TABLE I
FINAL MODEL QUALITY (ROUND 50, 5 SEEDS)

Metric	Hetero (global)	Homogeneous (Pima)
AUC	0.888 ± 0.005	0.787 ± 0.019
AUC — Diabetes	0.798 ± 0.013	—
AUC — Heart	0.845 ± 0.011	—
AUC — Cancer	0.996 ± 0.002	—
Sensitivity	0.638 ± 0.021	0.312 ± 0.067
Specificity	0.908 ± 0.009	0.900 ± 0.033
F1 Score	0.712 ± 0.011	0.412 ± 0.064
Accuracy	0.807 ± 0.004	0.696 ± 0.023

TABLE II
HETEROGENEOUS FEDERATION: LOCAL VS. FEDERATED AUC (ROUND 50, 5 SEEDS)

Disease	Local (isolated)	Federated	Δ AUC
Diabetes	0.799 ± 0.058	0.798 ± 0.012	-0.001
Heart	0.910 ± 0.030	0.845 ± 0.011	-0.065
Cancer	0.995 ± 0.003	0.996 ± 0.002	$+0.001$

Positive Δ = federation improves on local training. Diabetes variance reduction: $\pm 0.058 \to \pm 0.012$ (4.8×).

To determine whether cross-disease gradient sharing provides utility, we compare against a *local baseline* in which each hospital trains in complete isolation for the same total gradient steps (50 rounds × 5 local epochs). Table II shows the results. Cancer (WDBC, n=456 training) is self-sufficient: local 0.995 ± 0.003 vs. federated 0.996 ± 0.002 ($\Delta = +0.001$). Heart disease incurs a **gradient-dilution penalty**: federated AUC (0.845 ± 0.011) is 6.5 points below local (0.910 ± 0.030) because WDBC gradients from semantically disjoint features interfere with Cleveland's 13-dimensional signal. Diabetes shows the same mean (0.799 vs. 0.798) but federation reduces prediction variance **4.8**× ($\pm 0.058 \to \pm 0.012$), providing more stable clinical decision-support for hospitals with only ∼200 training samples.

These results reveal an inherent tradeoff in cross-disease federation: variance stabilisation for data-scarce nodes versus accuracy degradation from gradient dilution. Critically, the PQC security layer operates identically in both regimes; the quantum-safety guarantees are independent of whether the federation scope is clinically optimal. Determining appropriate federation scope is a domain decision; future work should explore multi-task FL architectures (per-disease output heads) to eliminate gradient interference while preserving variance reduction.

B. PQC vs. Classical Cryptography

Table III compares PQC (ML-KEM-768 + ML-DSA-65 + AES-256-GCM) against the classical baseline (RSA-4096 + HMAC-SHA256 + AES-256-GCM) over 30 rounds and 5 nodes. Both schemes reach identical model quality (AUC 0.837), confirming that replacing classical primitives with PQC does not affect convergence.

PQC is **1.6× faster** per round (64 ms vs. 101 ms). This advantage comes primarily from decryption: ML-KEM-768

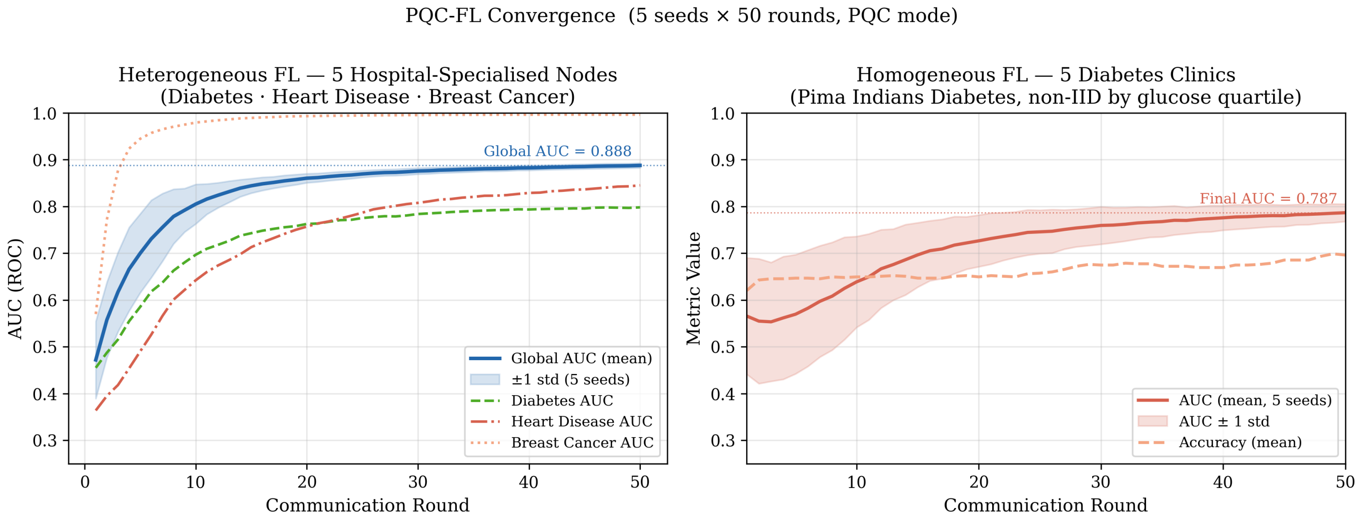

Fig. 1. FL convergence over 50 rounds across 5 random seeds. **Left:** Heterogeneous federation—each of the 5 hospitals specialises in one disease domain. Solid line: global AUC 0.888 ± 0.005; dashed lines show per-disease AUC (cancer 0.996, heart 0.845, diabetes 0.798). **Right:** Homogeneous federation (primary)—5 diabetes clinics partitioned non-IID by glucose quartile (AUC 0.787 ± 0.019). Shaded bands: ± 1 std across seeds.

TABLE III
PQC VS. CLASSICAL: 30 ROUNDS, 5 NODES

Metric	PQC	Classical
AUC	0.837	0.837
Avg round time (ms)	64	101
Avg enc / node (ms)	0.57	0.29
Avg dec / round (ms)	1.41	33.44
Total bandwidth (30r)	3381 KB	2531 KB
Crypto overhead / round	31.1 KB	2.8 KB

TABLE IV
NIST SECURITY LEVEL COMPARISON (12 KB PAYLOAD, 20 ITER.)

Metric	Level 1 ML-KEM-512 +ML-DSA-44	Level 3 ML-KEM-768 +ML-DSA-65	Level 5 ML-KEM-1024 +ML-DSA-87
Enc+Sign (ms)	0.72	0.41	0.46
Dec+Verify (ms)	0.22	0.31	0.31
Round-trip (ms)	0.94	0.71	0.77
KEM ciphertext (B)	768	1088	1568
Signature (B)	2420	3309	4627
Crypto overhead (KB)	4.4	6.2	8.6

decapsulation takes 1.41 ms per round, while RSA-4096 decryption takes 33.44 ms, a **24× speedup**. RSA decryption cost scales with key size, whereas lattice-based KEM decapsulation runs in $O(n \log n)$ polynomial ring arithmetic. Note that node-side PQC encrypt+sign (0.57 ms) is modestly slower than RSA encrypt+HMAC (0.29 ms); the end-to-end advantage is driven by server-side decapsulation, which dominates round time. Fig. 2 shows per-round latency traces across all 30 rounds.

C. NIST Security Level Comparison

Fig. 3 and Table IV benchmark all three NIST post-quantum security levels on a 12 KB model-delta payload (20 iterations each). Round-trip latency stays below 1 ms at all levels. Level 3 (ML-KEM-768 + ML-DSA-65) is slightly faster than Level 1 in our environment (0.713 ms vs. 0.940 ms), which we attribute to CPU cache behaviour and the optimisation profile of liboqs 0.15.0. Cryptographic overhead grows with security level: 4.4 KB (Level 1), 6.2 KB (Level 3), 8.6 KB (Level 5). For a 12 KB model delta, even the highest level adds 72% overhead in bytes, a modest cost for quantum-safe guarantees.

D. Byzantine Attack and Defence

Fig. 4 shows the two-part Byzantine evaluation over 20 rounds with node 4 as the rogue node.

Part A: Tampered Signature Attack. The rogue node flips the first 32 bytes of its ML-DSA signature before transmission. ML-DSA verification rejects the bundle in **100% of rounds (20/20)**. The global model trains on the remaining four honest nodes and converges more slowly than the clean baseline, reflecting the lost data contribution from one hospital shard, but no adversarial gradient ever enters the global model.

Part B: Gradient Poisoning Attack. The rogue node sends a correctly signed but 50×-scaled gradient delta. Without the norm filter, the poisoned model reaches AUC 0.567 at round 20, far below the clean baseline. With the gradient-norm filter (3× median threshold), the rogue node is rejected in **100% of rounds (20/20)**, and the defended model recovers to AUC 0.801. The delta between defended (0.801) and undefended (0.567) AUC directly illustrates the value of the second-layer filter. These results confirm that ML-DSA authentication provides a cryptographic first line of defence, and the norm filter forms an effective statistical second line.

E. Differential Privacy Tradeoff

Fig. 5 and Table V show the privacy-utility tradeoff across six epsilon values (30 rounds, seed 42, clip $S = 0.1$, $\delta = 10^{-5}$,

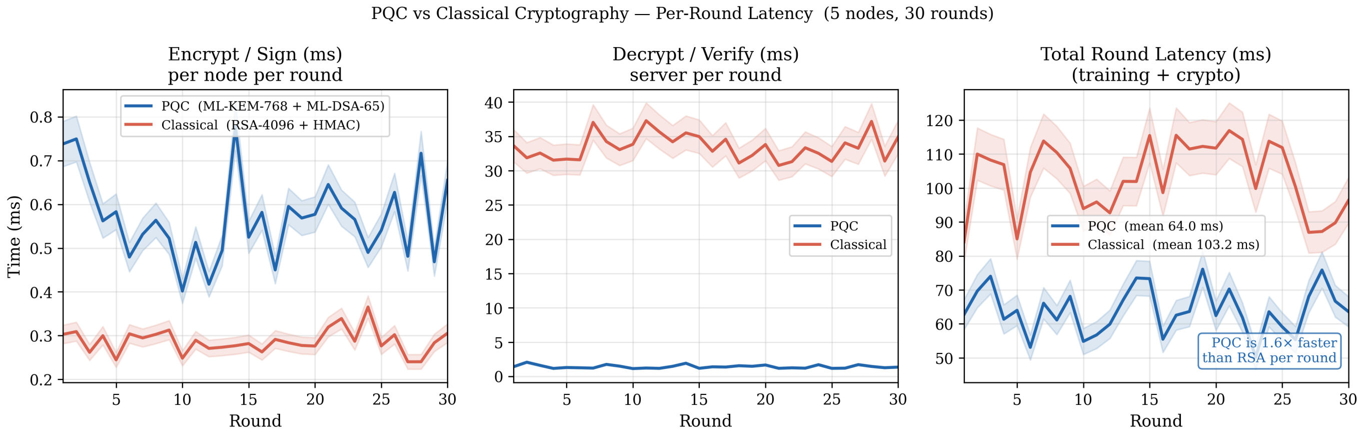

Fig. 2. Per-round latency breakdown over 30 rounds (5 nodes). **Left:** Node-side encrypt+sign time per round. **Centre:** Server-side decrypt+verify time per round. **Right:** Total round time (training + cryptography). PQC decryption (1.41 ms) is 24× faster than RSA-4096 (33.44 ms), yielding a 1.6× end-to-end speedup (64 ms vs. 101 ms).

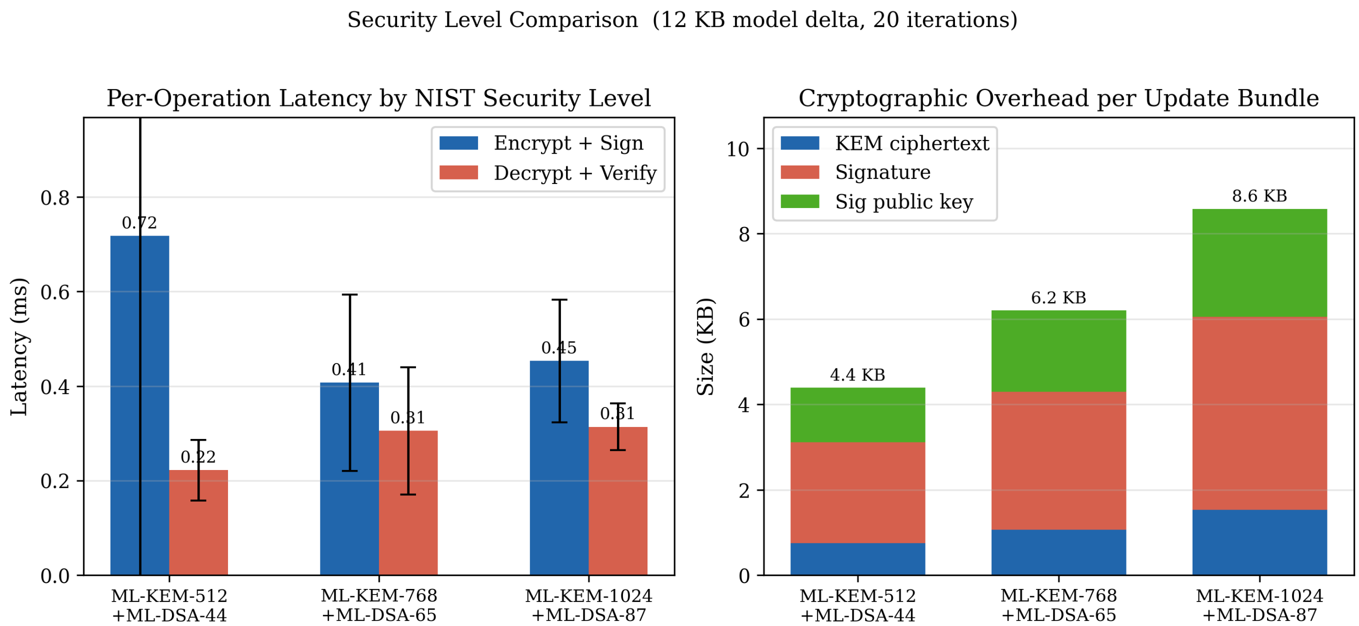

Fig. 3. NIST post-quantum security level comparison on a 12 KB model-delta payload (20 iterations each). **Left:** Per-operation latency (mean ± std). All levels complete encrypt+sign and decrypt+verify under 0.8 ms. **Right:** Cryptographic overhead per update bundle, broken down into KEM ciphertext, signature, and signing public key. Overhead ranges from 4.4 KB (Level 1) to 8.6 KB (Level 5).

lr $= 0.01$).

At $\varepsilon = \infty$ (no DP), the model reaches AUC 0.845. At $\varepsilon = 50$, the penalty is only -1.3% (AUC 0.834). At $\varepsilon = 10$, the penalty is -4.4% (AUC 0.808). Tighter budgets cause severe degradation: AUC 0.741 at $\varepsilon = 5$ and near-random performance at $\varepsilon \leq 1$.

This steep degradation at small ε is a known property of DP in small-model FL [8]: with only 1531 parameters, the Gaussian noise ($\sigma = 0.484$ at $\varepsilon = 1$) overwhelms the gradient signal ($\|\Delta\|_2 \approx 0.07$). At $\varepsilon = 0.5$ ($\sigma = 0.969$), the noise-to-signal ratio exceeds 13×, causing the sigmoid output to saturate and the model to produce non-discriminative outputs, yielding AUC 0.000. In this degenerate regime the model assigns lower predicted probability to positive-class inputs than to negative-class inputs—a complete inversion of the intended mapping—so our ROC integrator returns 0.000 (a perfectly anti-correlated classifier). The practical recommendation is that $\varepsilon \geq 10$ **provides meaningful privacy at acceptable utility cost** for this model class.

F. Communication Overhead

Table III quantifies per-round bandwidth. Each round, five nodes transmit bundles totalling 112.7 KB (PQC) vs. 84.4 KB (classical), with 81.6 KB model payload common to both. PQC overhead is 31.1 KB per round: ML-KEM ciphertexts

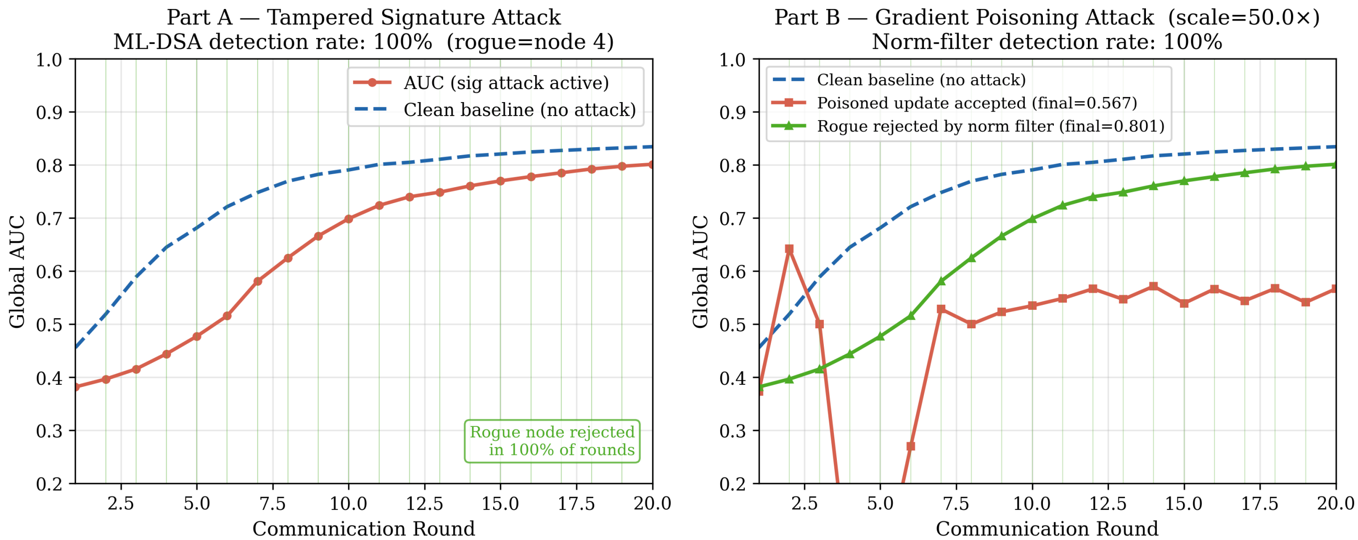

Fig. 4. Byzantine attack evaluation over 20 rounds (node 4 as rogue, seed 42). **Left (Part A):** Tampered-signature attack. ML-DSA rejects the rogue node in 100% of rounds. **Right (Part B):** Gradient-poisoning attack (50× scale). Without defence, final AUC drops to 0.567. With the gradient-norm filter, the rogue is rejected in 100% of rounds and AUC recovers to 0.801 (+0.235 delta).

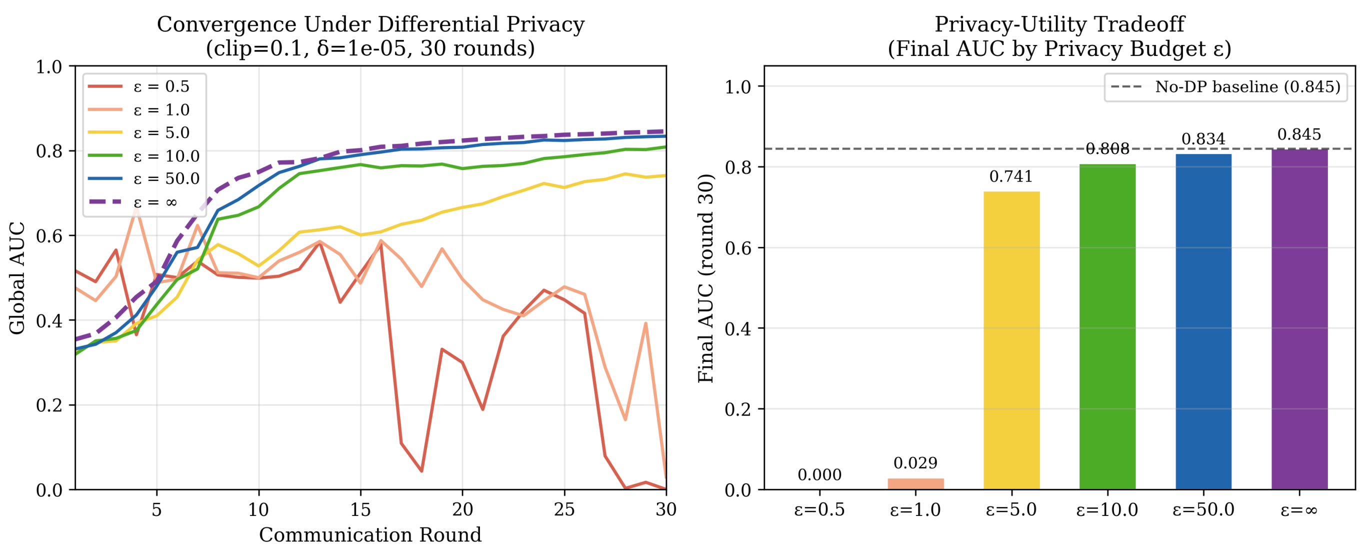

Fig. 5. Differential privacy tradeoff (30 rounds, seed 42, clip $S = 0.1$, $\delta = 10^{-5}$). **Left:** Convergence curves per ε value. $\varepsilon = 50$ nearly matches the no-DP baseline (dashed). **Right:** Final AUC at round 30 vs. privacy budget. At $\varepsilon = 10$, AUC penalty is only -4.4%. At $\varepsilon \leq 1$, noise overwhelms the gradient signal.

TABLE V
DIFFERENTIAL PRIVACY: FINAL AUC (ROUND 30) VS. ε

ε	σ	**AUC (Round 30)**
0.5	0.969	0.000
1.0	0.484	0.029
5.0	0.097	0.741
10.0	0.048	0.808
50.0	0.010	0.834
∞	0	0.845

(5.3 KB), ML-DSA-65 signatures (16.1 KB), and signing public keys (9.5 KB, amortisable as long-term keys registered once at setup). Over 30 rounds, PQC consumes 850 KB more bandwidth than classical—at 1 Mbps, about 6.8 additional seconds per training run, a negligible cost for quantum-safe guarantees.

G. Scalability

To test how PQC-FL-Health scales beyond the five-hospital baseline, we extend the federation to 10 and 20 simulated nodes, repeating the mixed-dataset protocol over three seeds

TABLE VI
PQC-FL-HEALTH SCALABILITY (50 ROUNDS, 3 SEEDS)

Nodes	Final AUC	Avg Round (ms)
5	0.862 ± 0.010	63.8 ± 1.0
10	0.862 ± 0.010	63.0 ± 0.2
20	0.862 ± 0.010	65.5 ± 0.8

AUC invariant by FedAvg construction (proportional partitioning); the key finding is flat PQC latency across all scales.

(50 rounds each). Table VI summarises final AUC and average round time for all three configurations.

PQC overhead does not compound with federation size. Average round time is 64 ms at 5 nodes, 63 ms at 10 nodes, and 65 ms at 20 nodes—a variation of under 3% across a $4\times$ increase in participants. This reflects the cost structure of ML-KEM and ML-DSA: encapsulation and signature verification are fixed-cost polynomial ring operations that do not compound with the number of participants. Final AUC is 0.862 ± 0.010 at all three scales; under proportional FedAvg partitioning, this is expected by construction (the weighted gradient average is invariant to the number of equal shards). The genuine scalability finding is the flat cryptographic overhead: quadrupling the federation produces no measurable latency penalty.

VI. CONCLUSION

We presented PQC-FL-Health, a federated learning system for healthcare that combines three complementary privacy guarantees: quantum-safe confidentiality via ML-KEM-768, authenticated gradient integrity via ML-DSA-65, and aggregator-side patient privacy via Gaussian differential privacy. On a node-specialised five-hospital heterogeneous federation spanning three disease domains, the system achieves global AUC 0.888 ± 0.005 over 50 rounds and 5 seeds, with per-disease AUC of 0.798 (diabetes), 0.845 (heart disease), and 0.996 (breast cancer). A local-vs.-federated comparison shows that cross-disease federation reduces prediction variance $4.8\times$ for small-dataset hospitals while introducing gradient-dilution penalties for heart disease, quantifying the tradeoff that practitioners must evaluate before deploying cross-domain federations. PQC operations complete in under 1 ms at all NIST security levels and run $1.6\times$ faster end-to-end than RSA-4096, despite transmitting $11\times$ more cryptographic bytes per update. ML-DSA verification achieves 100% detection on tampered-signature attacks, and the gradient-norm filter catches all tested scaled-gradient poisoning attempts (single rogue node, $50\times$ scale, seed 42). The DP layer shows that $\varepsilon \geq 10$ costs under 5% AUC for this model class. Scalability experiments confirm that round latency remains flat at 63–65 ms when the federation grows from 5 to 20 nodes, with PQC cryptographic overhead not compounding with federation size.

Current limitations include the small model size (1531 parameters, which restricts viable DP budgets), single-CPU experiments without HSM acceleration, single-rogue-node Byzantine scenarios, and simulated rather than real network conditions. Future work should explore multi-task FL architectures (per-disease output heads) to eliminate gradient dilution while preserving variance reduction, and test larger models where tight DP budgets are more viable.

As quantum hardware matures, federated healthcare systems should treat PQC as a present design requirement rather than a future upgrade. PQC-FL-Health shows that quantum-safe FL is practical today, with no accuracy cost and a latency advantage over the classical baseline it replaces.

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